

BT501 Current Papers Solved

#MidTerm

Spring -2022

We Grow Together ✨

What is Parkinson's disease?

Parkinson's disease is a brain disorder that causes unintended or uncontrollable movements, such as shaking, stiffness, and difficulty with balance and coordination. Symptoms usually begin gradually and worsen over time. As the disease progresses, people may have difficulty walking and talking.

Define Huntington disease?

Huntington's disease is a rare, inherited disease that causes the progressive breakdown (degeneration) of nerve cells in the brain. Huntington's disease has a wide impact on a person's functional abilities and usually results in movement, thinking (cognitive) and psychiatric disorders. Huntington's disease is caused by a faulty gene that results in parts of the brain becoming gradually damaged over time.

Define Basophil?

Basophils are a type of white blood cell. Basophils are the least common type of granulocyte, representing about 0.5% to 1% of circulating white blood cells. However, they are the largest type of granulocyte.

Types and names of immune disease?

- Hashimoto thyroiditis
- Celiac disease
- Graves' disease
- Addison's disease
- Inflammatory bowel disease
- Multiple sclerosis
- Rheumatoid arthritis
- Type 1 diabetes

Name of bio marker for ovarian carcinoma and Steps of sandwich ELISA?

Name of bio marker for ovarian carcinoma

CA125 is the best-known serum ovarian cancer biomarker, it is not the only one: CEA (mucinous), LDH (dysgerminoma, mixed germ cell tumors), β -hCG (choriocarcinoma, mixed germ cell tumors), inhibin B (granulosa cell tumors), α -fetoprotein (yolk sac tumors, embryonal cell tumors), and HE4 are also available.

Steps of sandwich ELISA

- Coat a surface with capture antibody.
- Block unbound protein binding sites on the surface
- Add the sample containing the target antigen
- Add the detection antibody label conjugate
- Detection

Human genome project?

The Human Genome Project (HGP) was an international scientific research project with the goal of determining the base pairs that make up human DNA, and of identifying, mapping and sequencing all of the genes of the human genome from both a physical and a functional standpoint. Goals of the human genome project include: Optimization of the data analysis. Sequencing the entire genome. Identification of the complete human genome.

What is Immunity?

Immunity is the ability of the body to defend itself against disease-causing organisms. Everyday our body comes in contact with several pathogens, but only a few results into diseases. The reason is, our body has the ability to release antibodies against these pathogens and protects the body against diseases. This defence mechanism is called immunity.

Types of Immunity

There are two major types of immunity:

Active immunity

Active immunity occurs when our own immune system is responsible for protecting us from a pathogen.

Passive immunity

Passive immunity occurs when we are protected from a pathogen by immunity gained from someone else.

Difference between innate and adaptive immunity system?

Innate immunity

Innate immunity refers to nonspecific defense mechanisms that come into play immediately or within hours of an antigen's appearance in the body. These mechanisms include physical barriers such as skin, chemicals in the blood, and immune system cells that attack foreign cells in the body. The innate immune response is activated by chemical properties of the antigen.

Adaptive immunity

Adaptive immunity refers to antigen-specific immune response. The adaptive immune response is more complex than the innate. The antigen first must be processed and recognized. Once an antigen has been recognized, the adaptive immune system creates an army of immune cells specifically designed to attack that antigen. Adaptive immunity also includes a "memory" that makes future responses against a specific antigen more efficient.

What does the red color indicate in DNA microarray?

A red spot indicates that that gene was strongly expressed in cancer cells.

What is metagenomics?

Metagenomics is the study of the structure and function of entire nucleotide sequences isolated and analyzed from all the organisms (typically microbes) in a bulk sample.

Different between orthologs and paralogs?

Orthologs

orthologs are defined as homologs in different species that catalyze the same reaction.

Paralogs

Paralogs are defined as homologs in the same species that do not catalyze the same reaction.

Which chromosome has maximum and minimum number of gene?

In the human genome, chromosome 1 has the most genes. It has 2968 genes. Chromosome Y contains the least number of genes.

What is Xist activity regulatory?

Xist (X-inactive specific transcript) is a non-coding RNA on the X chromosome of the placental mammals that acts as a major effector of the X-inactivation process. It is a component of the Xic – X-chromosome inactivation centre along with two other RNA genes and two protein genes.

Structure of the following types of viruses e. g. Single strand or double?

Adenovirus

Double-stranded DNA genome

Hepatitis B

Partially double-stranded DNA

Polio virus

Single-stranded RNA

Influenza virus

Single stranded RNA

What is DNA microarray?

The DNA microarray is a tool used to determine whether the DNA from a particular individual contains a mutation in genes like BRCA1 and BRCA2. The chip consists of a small glass plate encased in plastic.

Note on Cloning Vectors?

Cloning Vector

“A cloning vector is a small piece of DNA into which a foreign DNA can be inserted for cloning purposes.” Cloning vectors are utilized to insert foreign DNA into another cell and create multiple copies of the same. The foreign DNA is duplicated and expressed utilizing the host cell machinery. It amplifies one copy of DNA into multiple copies.

Features of Cloning Vectors

The cloning vectors possess the following features:

- A cloning vector should possess an origin of replication so that it can self-replicate inside the host cell.
- It should have a restriction site for the insertion of the target DNA.
- It should have a selectable marker with an antibiotic resistance gene that facilitates screening of the recombinant organism.
- It should be small in size so that it can easily integrate into the host cell.
- It should be capable of inserting a large segment of DNA.
- It should possess multiple cloning sites.
- It should be capable of working under the prokaryotic and eukaryotic systems.

Types of Cloning Vectors

There are the following different types of cloning vectors:

- Plasmids
- Bacteriophage
- Phagemids
- Bacterial Artificial Chromosomes

Advantages of PCR-based tests?

- Faster Results
- Shorter Time to Optimal Therapy
- Improve Treatment Decisions
- Avoid Unnecessary Antibiotics
- Reduce Unnecessary Testing
- Reduce Healthcare Costs

What is Real-time PCR or qPCR?

A real-time polymerase chain reaction is a laboratory technique of molecular biology based on the polymerase chain reaction. It monitors the amplification of a targeted DNA molecule during the PCR, not at its end, as in conventional PCR. Real-time PCR can be used quantitatively and semi-quantitatively.

Recombinant DNA technology and its rDNA cloning steps?

Recombinant DNA technology

Recombinant DNA technology is the joining together of DNA molecules from two different species. The recombined DNA molecule is inserted into a host organism to produce new genetic combinations that are of value to science, medicine, agriculture, and industry.

Steps of Recombinant DNA technology

Step-1. Isolation of Genetic Material.

Step-2. Cutting the gene at the recognition sites.

Step-3. Amplifying the gene copies through Polymerase chain reaction (PCR).

Step-4. Ligation of DNA Molecules.

Step-5. Insertion of Recombinant DNA Into Host.

Point independent method of mapping for diseased gene?

In Gene mapping Recombination frequency and gene distance in the chromosome are directly proportional. On one chromosome, there is a different genes present at a different distance. So there is a direct relation between a gene distance and a recombination frequency. For disease it is important and measured by the linear order by genetic mapping.